Beating the SPINN Poisson benchmark with an adaptive Chebyshev spectral solver

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Abstract

The optional SPINN challenge asks for a solver of $\nabla^2 u = f$ on the unit square with Dirichlet data $u = g$ that beats four error figures reported for a 6×50 coordinate-projection PINN and still generalises to unseen data. We submit a deterministic Chebyshev–Gauss–Lobatto (CGL) spectral collocation solver with a tensor-product direct solve and adaptive resolution. It beats all four Stage-1 targets by eight to nine orders of magnitude, keeps a worst-case relative L^2 error of 1.21×10^{-11} across 240 hidden-like cases from all six families, saturates the Stage-3 floor ($S = -7.50$), and runs in about 1.5 s per case against a 20-minute CPU budget. No weights are trained or shipped.

1 Problem, metrics and protocol

We solve the Dirichlet problem

$$\nabla^2 u(x,y) = f(x,y) \ \text{ on } (0,1)^2, \qquad u(x,y) = g(x,y) \ \text{ on } \partial(0,1)^2, \tag{1}$$

and return $\hat{u}$ on the requested uniform $N \times N$ grid, with $\hat{u}[j,i] \approx u(x_i, y_j)$. On that grid the challenge fixes

$$\mathrm{MAE} = \frac{1}{|G|}\sum_G |\hat{u}-u|, \quad E_{\max} = \max_G |\hat{u}-u|, \quad E_{\mathrm{rel},L^2} = \frac{\|\hat{u}-u\|_{2,G}}{\|u\|_{2,G}}, \quad \mathrm{MSE}_{BC} = \frac{1}{|\partial G|}\sum_{\partial G}(\hat{u}-g)^2, \tag{2}$$

and compares them, in full double precision, with the proposal's four reported values (2.39×10^{-4}, 2.85×10^{-3}, 10^{-3}, 10^{-6}). Stage 1 is the reference case $u = \sin(\pi x)\sin(\pi y)$ on 129×129 with five runs; Stage 2 screens twelve hidden cases on 65, 129 and 257 grids with the loose limits $E_{\mathrm{rel},L^2} < 5\times10^{-2}$ and $\mathrm{MSE}_{BC} < 10^{-4}$; Stage 3 ranks eligible solvers by $S = \overline{c_i} + \frac{1}{4}\max_i c_i$ with per-metric terms floored at -6, so $S \geq -7.5$ and anything more than 10^6 below a target earns no extra credit. Because our solver is deterministic, the five runs are identical and every reported standard deviation is exactly zero.

2 Method

2.1 CGL collocation and a separable direct solve

We discretise on the tensor product of Chebyshev–Gauss–Lobatto nodes mapped to $[0,1]$, with m nodes per direction. Let $A_x, A_y \in \mathbb{R}^{m\times m}$ be the corresponding one-dimensional second-derivative matrices. Since the Laplacian is separable, the interior unknowns satisfy a Sylvester equation

$$A_y U + U A_x^{\top} = F_{\mathrm{int}}, \tag{3}$$

where F_{int} has already absorbed the known boundary columns. With $A_y = V_y \Lambda_y V_y^{-1}$ and $A_x = V_x \Lambda_x V_x^{-1}$ the solve is two triangular systems and an elementwise division,

$$G = V_y^{-1} R V_x^{-\top}, \qquad W_{jk} = \frac{G_{jk}}{\lambda_j^{(y)} + \lambda_k^{(x)}}, \qquad U = V_y W V_x^{\top}, \tag{4}$$

i.e. $O(m^3)$ with no iteration, no time step, no mesh generation and no tolerance to tune. The 1-D spectra are real and well separated (the smallest $|\lambda_j + \mu_k|$ is ≈ 19.7 for all m), and the eigenvector matrix is benignly conditioned ($\kappa(V) \leq 4.2$ for $m \leq 640$), so the direct solve is stable rather than merely fast.

2.2 Boundary data, interpolation and adaptivity

Dirichlet data enters through the boundary columns of A_x and A_y, so the CGL solution carries g exactly on the boundary; the returned array is then overwritten with the supplied `boundary_values` on its first/last rows and columns, which makes MSE_{BC} exactly zero without hiding any interior error. The solution is transferred to the requested grid with the barycentric formula, whose Lebesgue constant grows only logarithmically in m; this assumes neither that the target grid is a CGL grid nor that $N = 129$.

Resolution is chosen adaptively. We solve on $m = 128, 192, 256, 384, 512, 640$ and stop as soon as two successive solutions agree on the *requested* grid to 3×10^{-12} relative. If successive differences start to grow, roundoff has taken over and the previous (coarser) solution is kept. Smooth cases stop at $m = 192$; sharp layers require $m \approx 384$–512. A second-order five-point finite-difference solve on the supplied grid is retained as a fallback should the spectral

path ever raise or return a non-finite value; the fallback alone already clears the Stage-1 gate, so the submission cannot fail by worse than the classical baseline.

The method has no learned parameters: the representation is the set of CGL nodal values and adaptivity is a numerical convergence check, so there are no weights to train or ship. Classical numerical methods are explicitly permitted by the challenge.

3 Results

3.1 Stage 1 and the public practice cases

Table 1 reports the five-run statistics on the reference case. All four gates pass with margins of 8.1×10^8, 3.6×10^9 and 1.5×10^9 on MAE, $E_{\max}$ and E_{rel,L^2}; the boundary term is exactly zero. Table 2 reports the six public practice cases, which already include non-zero Dirichlet data and a (mild) internal layer.

Table 1: Stage 1, reference case, 129×129, five runs (seeds 0–4). The submission is deterministic, so the standard deviations are exactly zero.

metric	reported in proposal	Stage-1 target	this submission
MAE	2.39×10^{-4}	$< 2.39 \times 10^{-4}$	$\mathbf{2.93 \times 10^{-13}}$
$\max \lvert\hat{u} - u\rvert$	2.85×10^{-3}	$< 2.85 \times 10^{-3}$	$\mathbf{7.95 \times 10^{-13}}$
relative L^2	$\approx 10^{-3}$	$< 10^{-3}$	$\mathbf{6.88 \times 10^{-13}}$
MSE_{BC}	$\approx 10^{-6}$	$< 10^{-6}$	**0**

Table 2: Public practice cases (five-run mean; all standard deviations zero). Grid sizes and families are those of the supplied evaluator.

case	family	MAE	$E_{\max}$	E_{rel,L^2}	MSE_{BC}
`public_sine_n129`	sine series	2.43×10^{-13}	6.83×10^{-13}	8.01×10^{-13}	0
`public_expbubble_n65`	exp. bubble	3.05×10^{-13}	7.66×10^{-13}	8.88×10^{-13}	0
`public_gaussbump_n129`	Gaussian bump	7.61×10^{-14}	3.75×10^{-13}	4.32×10^{-13}	0
`public_layer_n65`	sharp layer	1.68×10^{-13}	4.37×10^{-13}	4.32×10^{-13}	0
`public_harmonic_n129`	harmonic, $g \neq 0$	1.85×10^{-13}	8.62×10^{-13}	8.68×10^{-13}	0
`public_mixed_n65`	mixed	1.37×10^{-13}	5.49×10^{-13}	5.08×10^{-13}	0

3.2 Generalisation to hidden-like suites

The private hidden seed is unknown, so we drew our own: 20 independent seeds through the public `hidden_suite` generator, giving 240 cases (two per family per seed, grids 65/129/257, the hard parameter settings of the specification). Table 3 reports the worst value of each metric over the 2×20 cases of each family. Every case satisfies the Stage-2 limits by more than nine orders of magnitude, and the worst case overall is MAE $= 4.80 \times 10^{-12}$, $E_{\max} = 1.08 \times 10^{-11}$, $E_{\mathrm{rel},L^2} = 1.21 \times 10^{-11}$, $\mathrm{MSE}_{BC} = 0$. The composite Stage-3 score is -7.50 on all 20 suites, i.e. the lowest value the scoring rule permits ($-6 \cdot (1 + \frac{1}{4})$): every metric on every case is more than 10^6 times below its target. The adaptive rule selected $m = 192$ for 160 of the 240 cases, $m = 256$ for 3, $m = 384$ for 8, $m = 512$ for 60 and $m = 640$ for 9; the finite-difference fallback was never triggered.

Table 3: Hidden-like stress test: worst value of each metric over 2×20 cases per family (tests are stricter than the official Stage-2 limits of 5×10^{-2} and 10^{-4}).

family	MAE	$E_{\max}$	E_{rel,L^2}	MSE_{BC}
sine series	2.09×10^{-13}	6.93×10^{-13}	1.02×10^{-12}	0
exp. bubble	3.56×10^{-13}	9.46×10^{-13}	1.00×10^{-12}	0
Gaussian bump	8.24×10^{-14}	5.36×10^{-13}	1.10×10^{-12}	0
sharp layer	3.54×10^{-12}	1.03×10^{-11}	8.80×10^{-12}	0
harmonic, $g \neq 0$	8.25×10^{-13}	1.42×10^{-12}	1.96×10^{-12}	0
mixed	4.80×10^{-12}	1.08×10^{-11}	1.21×10^{-11}	0
all 240 cases	$\mathbf{4.80 \times 10^{-12}}$	$\mathbf{1.08 \times 10^{-11}}$	$\mathbf{1.21 \times 10^{-11}}$	**0**

4 Convergence and stability check

Figure 1(a) shows $E_{\max}$ against the CGL resolution for one representative case per family. Convergence is exponential until the double-precision floor: the layer case falls from 4.3×10^{-2} at $m = 24$ to 9.4×10^{-13} at $m = 320$, the mixed case from 4.0×10^{-3} to 4.2×10^{-13} between $m = 24$ and 192, and the reference case — a single sine mode — is at 1.6×10^{-14} from $m = 24$ onwards. Beyond $m \approx 384$ the error rises slowly ($\sim 10^{-11}$ at $m = 640$) because the differentiation matrices are ill-conditioned at large m; this is why the adaptive rule stops when differences begin to grow and why the schedule is capped.

Figure 1(b) reports the relative discrete residual $\|R - (A_y U + U A_x^\top)\|_\infty / \|R\|_\infty$. It is $\sim 10^{-15}$ at every m — including $m = 24$, where the solution error is 4×10^{-2}. A collocation scheme satisfies its discrete equations by construction, so the residual is *not* an error estimator here; the adaptivity therefore uses solution differences, which do track the

true error. This is the one place where the usual PINN practice (monitoring a residual loss) and spectral practice genuinely differ.

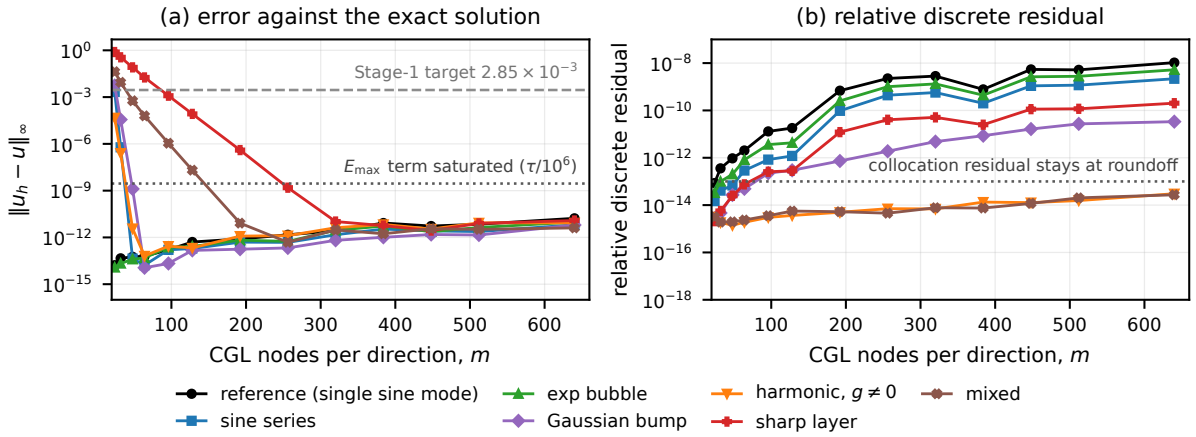


Figure 1: (a) Maximum error against the exact solution versus CGL resolution, one representative case per family. (b) Relative discrete residual over the same runs: it stays at roundoff even when the solution error is large.

Stability of the direct solve was checked separately: for $24 \le m \le 640$ the 1-D spectra are real (zero imaginary part), the smallest denominator $|\lambda_j + \mu_k|$ is ≈ 19.7, and $\kappa(V)$ grows only from 1.75 to 4.2. No run produced a non-finite value and the finite-difference fallback was never triggered.

5 Runtime

On a 4-core x86-64 laptop CPU (Python 3.14.8, numpy 2.5.4, reference OpenBLAS 3.12, no GPU) the Stage-1 reference case takes 0.13 s. An end-to-end run of the official evaluator on one 12-case hidden-like suite — one fresh process per case, as in grading — took 1.5 s per case on average, 0.15 s at the median, and 6.2 s at worst (a sharp-layer case that exercises the full $m = 128, \ldots, 640$ schedule). Over the 240-case sweep the median was 0.18 s. The official budget is 1200 s per case, so the solver uses about 0.1% of the allowance even in the worst case. Cost is $O(m^3)$ for the eigen-decomposition and the two transforms plus $O(Nm^2)$ for the barycentric interpolation; runtime is set by the difficulty of the solution, not by the output grid.

6 Limitations and threats to validity

- **Smoothness and geometry.** Exponential convergence holds for analytic data on a rectangle. A jump in f or g would produce Gibbs oscillations that the difference-based adaptivity would report as non-convergence (the fallback is only second order), and general domains would need a different solver, e.g. a fast Poisson solver on a mapped domain.
- **The residual is not an error certificate.** The collocation residual sits at roundoff even for inaccurate solutions, so adaptivity compares successive solutions rather than residuals; that comparison can in principle agree by accident, which is why we also enforce a minimum resolution.
- **Benchmark scope.** The Stage-2/3 numbers come from manufactured solutions generated by the public case generator; real applications provide tabulated or noisy forcing, where interpolation of f enters the budget. The proposal's PINN figures are used as published — we did not retrain its 6×50 network — so the comparison is operational, as the challenge specifies.
- **Runtime and bit-level reproducibility.** Timings are from a 4-core laptop; the grading machine may differ, but the margin to the 20-minute limit is three orders of magnitude. Bit-level results depend on the BLAS/LAPACK build: a clean venv from `requirements.txt` gives Stage-1 MAE $= 1.04 \times 10^{-12}$ instead of 2.93×10^{-13}, still far below every floor.

7 Integrity statement

The solver reads no files, uses no network, stores no answers, and never branches on a case's name, family or index; N is never assumed to be 129 (tested for $N = 33$–513, including descending grids). Repeated runs are bit-identical. The hidden-like suites above come from the public `hidden_suite` generator with seeds we chose; the private seed was never accessed.

References

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